

Moltwork - Google Ads API Design Document

Technical Design & Architecture

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1. Business Overview

Moltwork is an ecommerce intelligence platform that identifies product-market opportunities across European markets using automated research and controlled advertising experiments.

The platform focuses on finding products that are proven to sell in major markets (Germany, UK) but are poorly served in smaller affluent markets (Norway, Finland, Sweden, Denmark). We build lean, specialist stores for these gaps.

2. Tool Description

The Google Ads API tool is an internal research and campaign management system with three core functions:

- **Keyword Research Engine** - Automated queries to Keyword Planner API to measure search demand, advertiser competition, and cost-per-click estimates for specific products across target countries. This replaces manual Keyword Planner access and enables bulk analysis of hundreds of product-country combinations.
- **Competitor Intelligence** - Auction Insights queries to identify which competitors participate in Shopping auctions for our target product categories. This measures market saturation and identifies gaps where few competent sellers exist.
- **Campaign Management** - Automated Shopping campaign creation and management for validated product opportunities. Campaigns start at \$5-10/day as controlled experiments, with automated scaling rules based on performance metrics.

3. Architecture

The system is built as a modular Python pipeline:

Component	Description	API Used
Data Collection	Pull keyword volumes, CPC estimates, competition keywords	Keyword Planner API
Scoring Engine	Calculate opportunity scores using economic models and demand statistics	Internal Bayes API
Campaign Builder	Create Shopping campaigns for validated opportunities	Google Ads API
Performance Tracker	Monitor CTR, CPC, CVR, contribution profit per click	Google Ads API
Reporting	Generate performance reports and optimization recommendations	Google Ads API

4. Data Flow

1. Product candidates are identified through comparison engine analysis (Prisjakt, PriceRunner) and Merchant Center data.
2. For each candidate, the Keyword Planner API measures search demand and advertising cost in the target country.
3. The scoring engine calculates: break-even CVR = $CPC / (\text{selling price} - \text{all costs})$. Products with headroom > 1.3 proceed.
4. Validated products are added to Google Merchant Center via Merchant API.
5. Shopping campaigns are created at \$5-10/day with controlled product groups.
6. Performance data feeds back into the scoring model for learning.

5. Intended Users

Internal team only (2-3 people). No external users or client accounts are managed. The tool operates on a single Google Ads account (Customer ID: 3775149829).

6. API Usage Scope

API Feature	Usage	Volume
Keyword Planner	Keyword research for product opportunities	~500 queries/week
Auction Insights	Competitor analysis for target categories	~50 queries/week
Shopping Campaigns	Create/manage test campaigns	~5 campaigns/month
Reporting	Performance monitoring	Daily automated pulls
Customer Service	List accessible accounts	1 account (our own)

7. Compliance

- All API calls are made through official Google Ads Python client library (google-ads v31+).
- No scraping, no unauthorized data access, no third-party account management.
- Credentials stored in encrypted vault with access logging.
- Campaign budgets are hard-capped at \$10/day per experiment.
- All changes are logged with timestamps for audit trail.

8. Contact

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